TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	65.4 ± 2.87
Correlation Stability	82.3 ± 1.19
Distribution Consistency	73.4 ± 6.48
Entropy Preservation	88.0 ± 0.00
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	97.9 ± 1.14
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	97.3 ± 0.64
Novelty Score	1.0 ± 0.00

TabuLLM: AI Benchmark Report

Dataset Audit: breast_cancer_wisconsin_diagnostic.csv

1. Dataset Portrait

Attribute	Value
Rows	569
Columns	31
Numeric features	30
Categorical features	1
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
radius1	14.127	14.293	+0.166	1.2%	OK
texture1	19.290	19.400	+0.110	0.6%	OK
perimeter1	91.969	93.732	+1.763	1.9%	OK
area1	654.889	675.641	+20.752	3.2%	OK
smoothness1	0.096	0.096	-0.001	0.9%	OK
compactness1	0.104	0.107	+0.002	2.1%	OK
concavity1	0.089	0.094	+0.005	5.9%	OK
concave_points1	0.049	0.051	+0.002	4.3%	OK
symmetry1	0.181	0.182	+0.000	0.2%	OK
fractal_dimension1	0.063	0.063	+0.000	0.4%	OK
radius2	0.405	0.423	+0.018	4.4%	OK
texture2	1.217	1.211	-0.006	0.5%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	65.4 ± 2.87
Correlation Stability (*2)	82.3 ± 1.19
Distribution Consistency (*3)	73.4 ± 6.48
Entropy Preservation (*4)	88.0 ± 0.00
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	97.9 ± 1.14
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	97.3 ± 0.64
Novelty Score (*11)	1.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9587719298245613	0.9350877192982455	0.9535087719298245	-0.00526315789473684	
F1	0.9545900455578374	0.9213643782761881	0.9482883959590247	-0.00630164959881296	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9692982456140351	0.9359649122807017	0.9570175438596498	-0.01228070175438598	
F1	0.9657812227588884	0.9230873321797424	0.9528785745397184	-0.01290264821917021	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9394736842105263	0.9333333333333333	0.9377192982456141	-0.00175438596491228 62	
F1	0.9314283604116784	0.9177636300785721	0.9304603916787026	-0.00096796873297556 59	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9280701754385964	0.917543859649123	0.9245614035087717	-0.00350877192982456 1	
F1	0.9186755065587434	0.9020610727800422	0.9176747563770593	-0.00100075018168406 24	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9482456140350877	0.9166666666666666	0.895614035087719 3	-0.0526315789473684	
F1	0.9414539907562919	0.8927026152639412	0.8900449898277627	-0.05140900092852909	

Model: LR

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9508771929824562	0.9298245614035088	0.9508771929824562	0	
F1	0.9452155271956949	0.9104763279849724	0.9459880265583974	0.00077249936270246 12	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9605263157894737	0.9394736842105263	0.9614035087719298	0.00087719298245613 2	
F1	0.9558356677158535	0.9263085536337069	0.9567769825282657	0.00094131481241221 95	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9429824561403507	0.9219298245614036	0.9359649122807016	-0.00701754385964913	
F1	0.934337828992506	0.9039861391359457	0.9274170787091544	-0.00692075028335165 1	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9333333333333333	0.9201754385964913	0.9298245614035089	-0.00350877192982456 1	
F1	0.9242527459470589	0.9014687035986994	0.9193270157800054	-0.00492573016705343 2	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9710526315789474	0.9245614035087721	0.9614035087719298	-0.00964912280701756 3	
F1	0.9676338620041433	0.907045334774238	0.9569620664077734	-0.01067179559636991 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.99	10.38	122.8	1001.0	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871
2	20.57	17.77	132.9	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667
3	19.69	21.25	130.0	1203.0	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999
4	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744
5	20.29	14.34	135.1	1297.0	0.1003	0.1328	0.198	0.1043	0.1809	0.05883
6	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613
7	18.25	19.98	119.6	1040.0	0.09463	0.109	0.1127	0.074	0.1794	0.05742
8	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451
9	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
10	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243
11	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697
12	15.78	17.89	103.6	781.0	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082
13	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
14	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338
15	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682
16	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077
17	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922
18	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356
19	19.81	22.15	130.0	1260.0	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395
20	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766

B.1 Audit Trace: Fold 1 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	10.03	13.93	64.73	246.3	0.08946	0.06232	0.02783	0.005664	0.1991	0.07255
2	15.05	19.11	94.21	684.5	0.09587	0.1768	0.09966	0.1377	0.2069	0.06243
3	10.8	15.7	68.26	334.2	0.08363	0.08642	0.04711	0.0177	0.2027	0.06222
4	12.75	18.17	80.98	462.9	0.1022	0.1289	0.1578	0.01313	0.1934	0.05835
5	11.71	21.7	77.42	408.8	0.08491	0.08836	0.04505	0.05603	0.1649	0.05593
6	19.4	23.94	122.1	1138.0	0.1061	0.1555	0.1128	0.09464	0.1773	0.07102
7	13.24	20.13	86.87	542.9	0.08284	0.1223	0.101	0.02833	0.1601	0.06432
8	11.2	29.37	70.67	386.0	0.07449	0.03558	0.0	0.0	0.106	0.05502
9	16.74	21.59	110.1	869.5	0.0961	0.1336	0.1348	0.06018	0.1896	0.05656
10	13.17	18.66	85.98	534.6	0.1158	0.1231	0.1226	0.0734	0.2128	0.06777

B.2 Audit Trace: Fold 2 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	18.61	20.22	116.1	990.0	0.09469	0.1267	0.1478	0.04079	0.2166	0.06249
2	10.8	18.7	71.24	360.5	0.1044	0.1334	0.1594	0.1242	0.2197	0.06211
3	15.73	15.9	98.78	832.9	0.09714	0.09182	0.08092	0.04846	0.1922	0.06344
4	12.25	20.31	78.83	501.3	0.1063	0.04626	0.0	0.004167	0.1869	0.06894
5	12.2	15.46	85.84	462.0	0.08365	0.06136	0.0594	0.03132	0.1726	0.07325
6	14.2	25.13	93.97	609.9	0.1139	0.08468	0.145	0.09711	0.2556	0.05587
7	13.28	20.28	87.32	545.2	0.1041	0.1436	0.09847	0.06158	0.1974	0.06782
8	20.59	21.24	137.8	1320.0	0.1085	0.1644	0.2188	0.1121	0.1848	0.06222
9	12.18	20.52	77.22	458.7	0.08013	0.04038	0.02383	0.0177	0.1739	0.05677
10	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697

B.3 Audit Trace: Fold 3 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	19.59	16.85	158.9	1546.0	0.08946	0.09713	0.09875	0.06553	0.2127	0.06501
2	13.64	21.02	87.88	571.0	0.1089	0.1014	0.0839	0.04335	0.155	0.08261
3	11.13	18.7	71.8	360.5	0.09056	0.0363	0.004473	0.006423	0.192	0.05975
4	10.26	21.01	58.79	313.1	0.0778	0.1185	0.04591	0.03251	0.2026	0.06552
5	14.03	11.79	87.16	565.4	0.07351	0.1052	0.07097	0.0335	0.1788	0.05597
6	12.46	22.22	81.09	493.1	0.08268	0.1071	0.08448	0.0307	0.1621	0.05325
7	10.97	17.2	71.73	371.5	0.08915	0.1113	0.09457	0.03613	0.1489	0.0664
8	17.29	22.13	114.4	947.8	0.08999	0.1273	0.09697	0.07507	0.2108	0.05464
9	20.31	27.06	132.9	1288.0	0.1	0.1088	0.1519	0.09333	0.1814	0.05572
10	11.9	14.65	78.11	432.8	0.1152	0.1296	0.0371	0.03003	0.1995	0.07839

B.4 Audit Trace: Fold 4 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	15.34	16.99	100.2	716.6	0.1134	0.1066	0.169	0.141	0.1908	0.07682
2	14.19	16.85	93.77	646.1	0.08877	0.1113	0.05539	0.0378	0.1718	0.05677
3	11.37	18.47	75.03	435.6	0.09667	0.08836	0.06664	0.06527	0.2035	0.0611
4	15.75	20.25	97.65	758.6	0.1012	0.1188	0.1466	0.09702	0.1422	0.06088
5	12.34	17.31	77.88	462.9	0.08791	0.05223	0.01982	0.008829	0.1761	0.0652
6	19.59	21.84	130.4	912.7	0.08098	0.0763	0.1683	0.06638	0.1628	0.06782
7	20.09	23.86	134.7	1247.0	0.108	0.1838	0.2283	0.128	0.2249	0.07469
8	11.52	18.75	73.34	409.0	0.09524	0.05473	0.03036	0.02278	0.192	0.05907
9	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
10	17.75	28.03	117.3	981.6	0.09997	0.1314	0.1698	0.08293	0.1713	0.05916

B.5 Audit Trace: Fold 5 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	11.2	15.15	72.17	372.7	0.116	0.04038	0.01063	0.007583	0.1692	0.06899
2	22.01	17.46	134.7	1761.0	0.09231	0.1096	0.1122	0.06847	0.1853	0.05425
3	15.22	20.52	103.7	815.8	0.0925	0.1887	0.2448	0.1255	0.1943	0.05835
4	13.17	15.46	77.88	507.6	0.07557	0.04971	0.02772	0.02647	0.1737	0.06079
5	15.78	23.2	106.9	725.5	0.1044	0.2665	0.203	0.08683	0.1621	0.06684
6	15.75	19.66	110.1	773.5	0.07948	0.06712	0.07293	0.05669	0.1428	0.06069
7	8.726	15.83	55.84	230.9	0.115	0.08201	0.04132	0.01924	0.1649	0.07633
8	19.4	18.18	127.2	1145.0	0.1037	0.1442	0.1626	0.09464	0.1893	0.05892
9	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697
10	12.19	13.29	79.08	455.8	0.1066	0.09509	0.02855	0.02882	0.188	0.06471

B.6 Audit Trace: Fold 6 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	8.598	14.07	58.79	324.2	0.08855	0.1017	0.05077	0.02833	0.1801	0.06341
2	18.03	18.22	122.8	1094.0	0.09311	0.1516	0.1948	0.1377	0.2275	0.05335
3	11.37	19.07	70.79	366.5	0.1006	0.05223	0.01583	0.02036	0.1663	0.06243
4	12.46	18.83	79.83	388.1	0.1158	0.1262	0.1145	0.09194	0.1991	0.06113
5	16.25	18.8	97.83	761.7	0.09989	0.2084	0.1659	0.05364	0.1596	0.05935
6	12.89	20.74	79.01	538.4	0.09405	0.1317	0.06015	0.05596	0.1473	0.06309
7	11.6	12.84	74.34	412.6	0.08983	0.07525	0.04196	0.0335	0.162	0.06582
8	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
9	15.46	11.89	102.5	736.9	0.1257	0.1555	0.2032	0.1097	0.1966	0.07069
10	14.9	22.53	102.1	685.0	0.09947	0.2225	0.2733	0.09711	0.2041	0.06898

B.7 Audit Trace: Fold 7 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	11.5	19.63	76.83	420.5	0.1029	0.1305	0.09697	0.06158	0.1917	0.07469
2	11.41	20.54	73.88	386.0	0.08276	0.166	0.1937	0.1149	0.1773	0.05913
3	19.8	16.54	141.3	1264.0	0.0925	0.2084	0.2077	0.1118	0.2597	0.07029
4	17.29	23.95	123.6	1088.0	0.1065	0.07948	0.08169	0.02995	0.1589	0.07115
5	14.9	15.7	96.42	671.4	0.08217	0.1364	0.198	0.06606	0.1879	0.06284
6	13.71	20.97	86.49	562.1	0.1007	0.09218	0.03416	0.02315	0.1781	0.05955
7	19.89	20.26	130.5	1214.0	0.1037	0.131	0.1411	0.09431	0.1802	0.06188
8	17.42	25.56	114.5	948.0	0.1006	0.1146	0.1682	0.06597	0.1308	0.05866
9	11.95	14.96	77.23	426.7	0.1158	0.1206	0.01171	0.01787	0.2459	0.06581
10	14.69	13.98	98.22	656.1	0.1031	0.1836	0.145	0.063	0.2086	0.07406

B.8 Audit Trace: Fold 8 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.9	18.22	88.12	531.5	0.09834	0.1145	0.01462	0.02652	0.1521	0.07192
2	11.04	19.86	71.9	361.6	0.09597	0.04605	0.001461	0.007583	0.1893	0.06275
3	11.66	25.27	76.84	406.4	0.1036	0.08511	0.02987	0.0226	0.1621	0.07005
4	13.05	16.85	83.97	541.8	0.1128	0.0778	0.08817	0.05598	0.1467	0.05961
5	13.27	27.61	84.88	551.7	0.07551	0.109	0.03844	0.03876	0.1593	0.0632
6	19.55	19.98	130.0	1033.0	0.107	0.2363	0.1722	0.0852	0.1896	0.04996
7	13.05	18.59	85.09	512.0	0.1082	0.1304	0.09603	0.05603	0.2035	0.06501
8	10.16	19.59	64.73	311.7	0.1003	0.07504	0.005025	0.01116	0.1791	0.06331
9	16.24	18.77	108.8	805.1	0.1066	0.1802	0.1948	0.09052	0.1876	0.06684
10	14.87	20.21	96.12	680.9	0.09587	0.08345	0.06824	0.04951	0.1487	0.05748

B.9 Audit Trace: Fold 9 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	13.46	20.53	87.02	556.7	0.08365	0.09242	0.02974	0.02864	0.2196	0.05448
2	9.777	15.15	64.12	280.5	0.1128	0.07232	0.03	0.03279	0.19	0.06744
3	12.06	19.08	78.29	493.1	0.08853	0.06685	0.04132	0.01692	0.1943	0.05828
4	13.01	19.29	86.1	537.3	0.1109	0.1442	0.1411	0.0539	0.194	0.06216
5	11.95	20.67	73.53	432.7	0.09289	0.1141	0.04515	0.02757	0.1875	0.07696
6	20.64	17.88	134.8	1245.0	0.08876	0.1052	0.101	0.04603	0.194	0.07369
7	14.97	16.95	96.22	685.9	0.09855	0.07885	0.02602	0.03781	0.178	0.0565
8	11.43	15.39	73.06	399.8	0.09639	0.06889	0.03503	0.02875	0.1734	0.05865
9	11.81	17.39	75.27	428.9	0.1007	0.05562	0.02353	0.01553	0.1718	0.0578
10	10.57	18.32	66.82	340.9	0.08142	0.04462	0.01993	0.01111	0.2372	0.05768

B.10 Audit Trace: Fold 10 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.54	19.82	126.2	475.9	0.1001	0.128	0.1354	0.07041	0.1573	0.06432
2	9.683	17.25	60.21	245.2	0.08268	0.04994	0.006643	0.0151	0.1829	0.05278
3	11.68	28.06	76.37	416.2	0.08151	0.1141	0.01997	0.02037	0.1571	0.06612
4	14.99	23.77	97.83	668.7	0.08013	0.07773	0.1168	0.08271	0.1993	0.07237
5	9.504	16.15	64.41	248.7	0.06883	0.06601	0.05441	0.02471	0.2419	0.06161
6	16.84	21.35	93.97	928.2	0.1167	0.03872	0.0	0.005159	0.1618	0.06782
7	13.71	18.68	88.73	571.0	0.09916	0.107	0.05385	0.03783	0.1714	0.06843
8	11.94	18.24	75.71	437.6	0.08261	0.04751	0.01972	0.01349	0.1868	0.0611
9	10.51	23.09	66.85	334.2	0.1015	0.06797	0.02495	0.01875	0.1695	0.06556
10	11.75	17.56	75.89	422.9	0.1073	0.09713	0.05282	0.0444	0.1598	0.06677